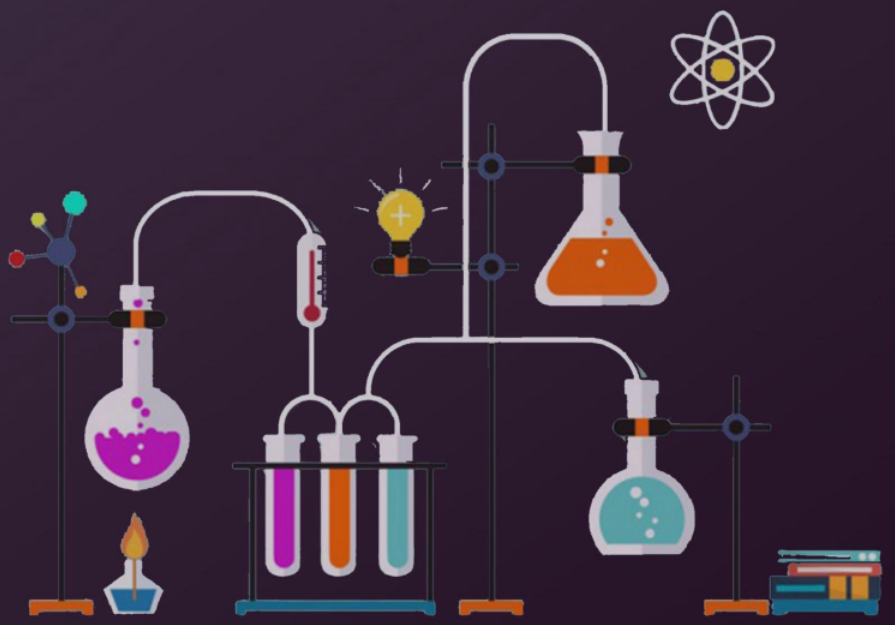
CHEMISTRY

Solid State



SHEET

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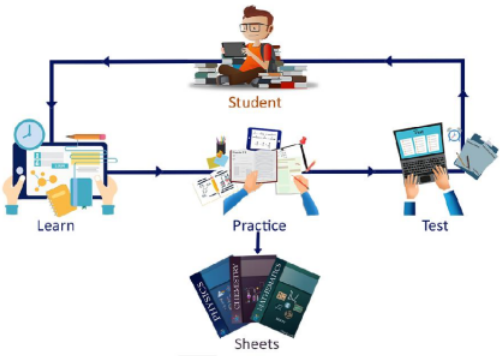
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction of Solid State	1			
2	Types of Solid	2-5	23,24,44,40	1	
3	Lattice and Unit Cells	6,7		2	
4	Bravais Lattices	8-10	1,2,25,26, 33-35,45	3,4	14,18-20
5	Coordination number and Packing Fraction	11-17	36, 46, 47,41	5-10	1,2,15
6	Packing in FCC and Density of Crystals	18-26	7,48-50	11,12	7-9
7	Close Packing in 1D, 2D and Square Close Packing in 3D	27-29	6,8-11	13-16	10-12
8	Hcp, Ccp and Body centered Packing	30-35	3-5, 27-31,51-53		13
9	Octahedral and Tetrahedral Voids	36-42	12,13	17-23	22
10	Packing in Ionic Solids	43-46	14,32,54	24-27	
11	Structures of Ionic Solids	47-55	15-18,37, 38,55	28-30	3-6,16,17
12	Imperfections in Solids, Frenkel and Schottky Defects	56-63	19-21,42,43	31	21
13	Electrical Properties of Solids		22,39		
14	Magnetic Properties of Solids	64			
15	ACC			32	

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Solid State

Q1. Which of the following is not the characteristic of solid :-
(A) fixed shape
(B) fixed volume
(C) low thermal energy
(D) low density

Types of Solid

Q2. A pseudo solid is :
(A) glass
(B) pitch
(C) KCl
(D) Glass and pitch both

Q3. Crystalline solids are generally in nature :-
(A) Rigid (B) Flexible
(C) Soft (D) None of these

Q4. Crystalline solids are?
(A) Anisotropic (B) Isotropic
(C) Both A and B (D) None of these

Q5. Amorphous solids may be classified as :-
(A) isotropic and super-cooled liquids
(B) anisotropic and super-cooled liquids
(C) isoenthalpic and super-heated liquids
(D) isotropic and super-heated solids.

Lattice and Unit Cell

Q6. The three dimensional graph of lattice points which sets the pattern for the whole lattice is called:-
(A) Space lattice (B) Simple lattice
(C) Unit cell (D) Crystal lattice

Q7. The number of atoms per unit cell in a simple cubic, face-centred cubic and body-centered cubic structure respectively are -
(A) 1, 4, 2 (B) 1, 2, 4
(C) 8, 14, 9 (D) 8, 4, 2

Bravais Lattices

Q8. Which of the following are the correct axial distances and axial angles for rhombohedral system?
(A) $a = b = c, \alpha = \beta = \gamma \neq 90°$
(B) $a = b \neq c, \alpha = \beta = \gamma = 90°$
(C) $a \neq b \neq c, \alpha = \beta = \gamma = 90°$
(D) $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90°$

Q9. Which of the following crystal system has more than one bravais lattice -
(A) Lattice with unit cell triclinic
(B) Lattice with unit cell hexagonal
(C) Lattice with unit cell monoclinic
(D) Lattice with unit cell rhombohedral

Q10. $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90°$ represents.
(A) tetragonal system
(B) orthorhombic system
(C) monoclinic system
(D) triclinic system

Coordination Number And Packing Fraction

Q11. In a face centred cubic cell, an atom at the corner contributes to the unit cell,
(A) 1 part (B) 1/2 part
(C) 1/4 part (D) 1/8 part

Q12. The coordination number of hexagonal closest packed(hcp) structure is
(A) 12 (B) 10
(C) 8 (D) 6

Q13. The space occupied in bcc arrangement is -
(A) 74% (B) 70%
(C) 68% (D) 60.4%

Q14. Lithium crystallizes as body centered cubic crystals. If the length of the side of unit cell is 350 pm, the atomic radius of lithium is :-
(A) 303.1 pm (B) 606.2 pm
(C) 151.5 pm (D) 123.7 pm

Q15. Body centred cubic and face centred cubic unit cells have n_1 and n_2 effective number of atoms respectively. Which one of the following $[n_1, n_2]$ combination is correct:
(A) [4, 1] (B) [4, 2]
(C) [1, 4] (D) [2, 4]

Q16. Which one of the following has highest packing efficiency ?
(A) Simple cubic
(B) Body centered cubic
(C) Hexagonal closed packing
(D) All have equal

Q17. $d = \frac{Z \times M}{N_A . a^3}$ $d \alpha z$ (Keeping other parameter fix)
Simple cubic Z = 1; B.C.C Z = 2
Hexagonal close packing Z = 6
Al (at wt. 27) crystallizes in the body centred cubic system with a = 4.05 Å.
Calculate the diameter of Al atom :-
(A) 1.732Å (B) 3.507Å
(C) 0.825Å (D) 1.650Å

Packing in FCC and Density of Crystals

Q18. An element (atomic mass = 60) having face centred cubic crystal has a density of 6.23 g cm^{-3}. What is the edge length of the unit cell
(Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1})
(A) 3×10^{-8} (B) 4×10^{-8}
(C) 6×10^{-8} (D) 2×10^{-8}

Q19. In a face centred cubic arrangement of A and B atoms where A are present at the corner and B at the face centres, A atoms are missing from 4 corners in each unit cell. What is the simplest formula of the compound ?
(A) AB_3 (B) A_2B_2 (C) AB_6 (D) AB_4

Q20. An element crystallises into a structure which may be described by a cubic type of a unit cell having one atom on each corner of the cube and two atoms on one of its diagonals. If the volume of this unit cell is 24×10^{-24} cm^3 and the density of the element is 7.2 g cm^{-3}, calculate the number of atoms present in 200g of the element ?
(A) 3.472×10^{24} (B) 1.1513×10^{24}
(C) 10.46×10^{24} (D) 1.8069×10^{24}

Q21. The face diagonal of a cubic closed packed unit cell is 4 Å. What is the length?
(A) 2.85 Å (B) 2.83 Å
(C) 2.86 Å (D) 2.87 Å

Q22. Silver crystallises in FCC lattice. If edge length of the cell is 4.077×10^{-8} cm and density is 10.5 g cm^{-3}, calculate the atomic mass of silver.
(A) 107.18 (B) 107.14
(C) 107.10 (D) 107.17

Q23. An element crystallises in BCC structure. The edge length of its unit cell is 288 pm. If the density of the crystal is 7.2 g cm^{-3}, what is the atomic mass of the element ?
(A) 51.8 (B) 103.6 (C) 25.9 (D) 207.2

Q24. The available space occupied by spheres of equal size in three dimensions in both hcp and ccp arrangement is
(A) 74% (B) 70% (C) 60.4% (D) 52.4%

Q25. A compound formed by elements A and B crystallises in cubic structure, in which atoms of A are at the corners while that of B are at the face centre. The formula of the compound is?
(A) AB_3 (B) AB
(C) A_3B (D) none of these

Q26. Each edge of a cubic unit cell is 400 pm long. If atomic weight of the element is 120 and it's density is 6.25 g/cm^3. The crystal lattice is? ($N_A = 6\times10^{23}$)
(A) Primitive cell (B) body centered
(C) face centered (D) All of these

Close Packing In 1D, 2D and Square Close Packing in 3D

Q27. What are the number of atoms per unit cell and the number of nearst neighbours in a simple cubic structure :-
(A) 1, 6 (B) 4, 12 (C) 2, 8 (D) 2, 6

Q28. If 5 Rs. coins are aaranged in square 2D packing then the packing efficiency is -
(A) 22% (B) 78.54%
(C) 52% (D) 68%

Q29. If identical spheres are arranged, in 2D in such a way bottom row sphere fits in dipression of top row, then the packing efficiency and co-ordination number is :
(A) 60%, 6 (B) 90%, 6
(C) 60%, 4 (D) 90%, 4

HCP ,CCP and Body Centered Packing

Q30. If the radius of an atom of an element is 75 pm and the lattice is body-centred cubic, the edge of the unit cell will be?
(A) 32.475 pm (B) 173.41 pm
(C) 37.5 pm (D) 212.1 pm

Q31. In a body centred unit cell, one of the diagonals contains two extra atoms. Find the number of atoms present per unit cell ?
(A) 1 (B) 4 (C) 2 (D) 6

Q32. The hcp and ccp structure for a given element would be expected to have :-
(A) the same alignment of layers
(B) the same density
(C) the same packing fraction
(D) all the above

Q33. Iron (At. mass = 56) has the bcc crystal lattice structure. Edge length of its unit cell is 278 pm. Radius of iron atom would be :-
(A) 120 pm (B) 278 pm
(C) 150 pm (D) 210 pm

Q34. Which of the following layering pattern will have a void fraction of 0.260 ?
(A) ABCCBAABC (B) ABBAABBA
(C) ABCABCABC (D) ABCAABCA

Q35. In the body-centered cubic unit cell & face centered cubic unit cell, the radius of atom in terms of edge lengh(a) of the unit cell is respectively :
(A) $\frac{a}{2}, \frac{a}{2\sqrt{2}}$ (B) $\frac{a}{2\sqrt{2}}, \frac{\sqrt{3}a}{4}$
(C) $\frac{\sqrt{3}a}{4}, \frac{a}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}a}{2}, \frac{a}{2\sqrt{2}}$

Octahedral and Tetrahedal Voids

Q36. The number of octahedral sites in a cubical close packed array of N spheres is?
(A) N/2 (B) 2N
(C) N (D) 4N

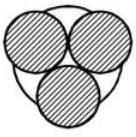
Q37. Atom A is every element of FCC, atom B is present at every Octahedral void, atom C is present at 25% of Tetrahedral void. Find out the possible molecular formula of the compound ?
(A) A_2B_2C (B) A_3B_3C
(C) A_4B_4C (D) A_5B_5C

Q38. The co-oridnation number of a cation occupying an octahedral hole is?
(A) 4 (B) 6 (C) 8 (D) 12

Q39. In a crystalline solid anion B are arranged in ccp lattice and cation A occupy 50% of the octahedral void and 50% of the tetrahedral voids. What is the formula of the solid ?
(A) A_3B_2 (B) AB
(C) A_2B_3 (D) A_4B_3

Q40. In a multi layered close-packed structure :
(A) there are twice as many tetrahedral holes as there are close-packed atoms
(B) there are as many tetrahedral holes as there are closed packed atoms
(C) there are twice as many octahedral holes as there are close-packed atoms
(D) there are as many tetrahedral holes as there are octahedral holes

Q41. The empty space between the shaded balls and hollow balls as shown in the diagram is called :-



(A) Hexagonal void
(B) octahedral void
(C) Tetrahedral void
(D) Double triangular void

Q42. If the anion (A) form hexagonal closest packing and cation (C) occupy only 2/3 octahedral voids in it, then the general formula of the compound is :-
(A) CA (B) CA_2
(C) C_2A_3 (D) C_3A_2

Packing in Ionic Solids

Q43. A solid A^+B^- has NaCl type close packed structure. If the anion has a radius of r_a 250 pm, what should be the ideal radius for the cation ? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal A^+B^- ?
(A) 56.25pm (B) 55.26pm
(C) 57.20pm (D) 56.20pm

Q44. If the radius of an octahedral void is r and the radius of the atom in close packing is R, derive the relation between r and R ?
(A) r = 0.412 R (B) r = 0.410 R
(C) r = 0.414 R (D) r = 0.418 R

Q45. In NaCl type structure shortest distance between two nearest neighbours is 100 pm, then distance between two next nearest neighbours in the same unit cell will be :
(A) $\frac{100}{\sqrt{2}}$pm (B) $100\sqrt{2}$pm
(C) 200 pm (D) $200\sqrt{2}$pm

Q46. The limiting radius ratio for tetrahedral shape is :-
(A) 0.414 to 0.732 (B) 0.225 to 0.414
(C) 0.155 to 0.225 (D) 0 to 0.155

Structures of Ionic Solids

Q47. Diamond belongs to the cyrstal system :
(A) Cubic (B) triclinic
(C) tetragonal (D) hexagonal

Q48. Which one of the following statements is incorrect about rock salt type :-
(A) It has FCC arrangement of Cl
(B) Na^+ and Cl^- ions have a coordination number of 6:6
(C) A unit cell of NaCl consists of four NaCl units
(D) All halides of alkali metals have rock salts type structure

Q49. The tetrahedral voids formed by ccp arrangement of Cl^- ions in rock salt structure are :
(A) Occupied by Na^+ ions
(B) Occupied by Cl^- ions
(C) Occupied by either Na^+ or Cl^- ions
(D) Vacant

Q50. The co-ordination number of Cs^+ and Cl^- ions in CsCl structure is :
(A) 4 : 4 (B) 6 : 6
(C) 8 : 8 (D) 4 : 8

Q51. Select the correct statement(s):-
(a) Co-ordination number of Cs^+ and Cl^- are 8 : 8 in CsCl crystal.
(b) For fcc lattice, radius of metal atom $= \frac{a}{2\sqrt{2}}$
(c) For triclinic system $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90°$
(d) In a fcc unit cell the body centre is an octrahedral vioid
Correct code is :-
(A) a, b, c (B) a, b, c, d
(C) b, c, d (D) a, c, d

Q52. Which one is the incorrect relation :-
(A) Number of tetrahedral voids = 2 × number of atoms present in closed placed arrengement
(B) Number of tetrahedral voids = number of octahedral voids
(C) Co-ordination number of Cs^+ in CsCl is 8
(D) Radius of octahedral voids = 0.414R

Q53. In the fluorite structure, the coordination number of Ca^{2+} ion is :-
(A) 4 (B) 6
(C) 8 (D) 3

Q54. Which of the following expressions is correct in case of a sodium chloride unit cell (edge lenght, a) ?
(A) $r_c + r_a = a$ (B) $r_c + r_a = a/2$
(C) $r_c + r_a = 2a$ (D) $r_c + r_a = \sqrt{2}a$

Q55. The unit cell of Caesium Chloride has one Cs^+ ion and
(A) One Cl^- (B) Two Cl^-
(C) Three Cl^- (D) Four Cl^-

Imperfections In Solids, Frenkel and Schottky Defects

Q56. In the Schottky defect
(A) cations are missing from the lattice sites and occupy the interstitial sites
(B) equal number of cations and anions are missing
(C) anions are missing and electrons are present in their place
(D) equal number of extra cations and electrons are present in the interstitial sites

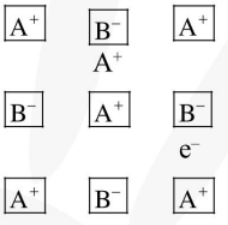
Q57. Ionic solids with Schottky defects contain in their structure,
(A) Equal number of cation and anion vacancies
(B) Anion vacancies and interstitial anions
(C) cation vacancies
(D) Cation vacancies and interstitial cations

Q58. At zero kelvin, most of the ionic crystals posses,
(A) Frenkel defect
(B) Schottky defect
(C) Metal excess deffect
(D) No defect

Q59. Frenkel defect is generally observed in,
(A) AgBr (B) Agl
(C) ZnS (D) All of these

Q60. In an ionic solid lattice,the cation has left a lattice site and is located at interstitial position, the lattice defect is :
(A) interstitial defect (B) vacancy defect
(C) Frenkel defect (D) Schottky defect

Q61. What type of crystal defect in indicated in the diagram below :-



(A) Frenkel defect (B) Schottky defect
(C) Both of them (D) None of these

Q62. Schottky defect causes :-
(A) Increase in the density of solid
(B) Decrease in the density of solid
(C) No change in the density of solid
(D) Decrease in the conductivity of solids

Q63. Select the incorrect statement -
(A) F-centres are a factor for imparting the colour to the crystal.
(B) Stoichiometry of crystal remains uneffected due to schottky defect.
(C) In orthorhombic crystal each interfacial angle is 90°.
(D) There are four unit cells in tetragonal crystal system.

Magnetic Properties of Solids

Q64. Anti ferromagnetism is present in :-
(A) ↑↑↑↑↑ (B) ↑↓↑↓
(C) Both the above (D) None

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following have two Bravais Lattice ?
(A) Cubic (B) Orthorhombic
(C) Tetragonal (D) Monoclinic

Q2. Which type of crystals contain one Bravais lattice ?
(A) hexagonal (B) triclinic
(C) rhombohedral (D) monoclinic

Q3. In a body centred unit lattice of A_2 type (Given : R = radius of atom A)
(A) the edge length is equal to $\frac{4R}{\sqrt{3}}$
(B) The edge length is equal to $2R^2 + 2R^2$
(C) The edge length is equal to $\frac{2d}{\sqrt{3}}$, where d = nearest neighbour distance
(D) The square of the edge length is equal to $16R^2$ where R is the radius of atom

Q4. The volume occupied by the atoms in F.C.C. unit cell are?
(A) $\frac{\pi}{6}r^3$ (B) $\frac{16}{3}\pi r^3$
(C) $\frac{16}{3}\pi^2 a^3$ (D) $\frac{\pi a^3}{3\sqrt{2}}$

Q5. A compound alloy of gold and copper crystallizes in a cubic lattice in which gold occupy the lattice point at corners of the cube and copper atom occupy the centres of each of the cube faces. What is the formula of this compound.
(A) $AuCu_6$ (B) AuCu
(C) $AuCu_3$ (D) $AuCu_2$

Q6. Diamond structure can be considered as ZnS (Zinc blend) structure in which each $Zn^{2+} = 4$ in alternate tetrahedral void and S^{2-} in cubic close pack arrangement is replaced by one carbon atom. If C–C covalent bond length in diamond is 1.5Å , what is the edge length of diamond unit cell (z = 8).
(A) 3.46 Å (B) 6.92 Å
(C) 1.73 Å (D) 3 Å

Q7. In a solid AB having NaCl type structure A atoms occupy the corner of the cubic unit cell and the face centered positions. If all the face-centered atom along one of the axes are removed, choose the correct options: (Atomic weight of A = 40), Atomic weight of B = 60, r_A = 1.4 Å, r_B = 2.8 Å
(A) The packing fraction is 0.579
(B) The density of the resultant solid is 1.008 g/cc
(C) The packing faction is 0.679
(D) The density of the resultant solid is 2.008 g/cc

Q8. Which of the following are correct with respect to body centred arrangement ?
(A) The spheres in the first layers of close packing are slightly opened up
(B) Each sphere in the scond layer is in contact with four spheres of the first layer and the third layer is similar to the first layer.
(C) The coordination number of the each sphere is 12.
(D) In bcc arrangement only 68% of the volume is occupied by the spheres.

Q9. KF crystallizes in the NaCl type structure. If the radius of K^+ ions 132 pm and that of F^- ion is 135 pm, what is the closet K–K distance ?
(A) Can not say (B) 534 pm
(C) 755 pm (D) 378 pm

Q10. Which one of the following schemes of ordering closed packed sheets of equal sized spheres generate close packed lattice.
(A) ABCABC (B) ABACABAC
(C) ABBAABBA (D) ABCBCABCBC

Q11. Select correct option/s
(A) ABABABAB.....in 3D is HCP
(B) ABCABCABC.....in 3D is CCP
(C) ABCABACABABC is close packing
(D) C.N. of both HCP and CCP atoms are 12

Q12. Select the correct statement about FCC (ABCAB...) structure.
(A) Distance between nearest octahedral void and tetrahedral void is $\frac{\sqrt{3}a}{4}$
(B) Distance between two nearest octahedral void is $\frac{a}{\sqrt{2}}$
(C) Distance between two nearest tetrahedral void is $\frac{\sqrt{3}a}{2}$
(D) Distance between layer A and B is $2r\sqrt{\frac{2}{3}}$

Q13. In the closest packing of atoms A (radius : r_a), the radius of atom B that can be fitted into tetrahedral voids is :-
(A) 0.155 r_a (B) 0.225 r_a
(C) 0.414 r_a (D) 0.732 r_a

Q14. Which of following is /are metallic solid.
(A) Fe (B) Mn (C) Ag (D) Zn

Q15. A cubic unit cell contains manganese ions at the corners and fluoride ions at the center of each edge. The radius of Mn ion is 0.65 Å and that of F^- ion is 1.36 Å.
(A) The empirical formula of compound is MnF_3
(B) The coordination number of the Mn ion is 6
(C) The edge length of the unit cell is 4.02 Å
(D) The density of the compound is 2.86 gm/cc

Q16. In A^+B^- ionic compound, radii of A^+ and B^- ions are 180 pm and 187 pm respectively. The crystal structure of this compound will be :-
(A) CsBr type (B) CsCl type
(C) ZnS type
(D) Similar to diamond

Q17. The melting point of RbBr is 682°C, while that of NaF is 988°C. The principal reason that melting point of NaF is much higher than that of RbBr is that –
(A) The two crystals are not isomorphous
(B) The molar mass of NaF is smaller than that of RbBr
(C) The internuclear distance ($r_c + r_a$) is greater for RbBr than that for NaF.
(D) NaF has greater lattice energy than that of RbBr

Q18. The mass of a unit cell of CsCl corresponds to :-
(A) 1 Cs^+ and 1 Cl^- (B) 1 Cs^+ and 6 Cl^-
(C) 4 Cs^+ and 4 Cl^- (D) 8 Cs^+ and 1 Cl^-

Q19. Choose the correct option/options -
(A) Two adjacent face centre atom doesn't touch each other in fcc unit cell because they are not nearest atom of face each other in fcc lattice.
(B) Number of nearest Na^+ ions of another Na^+ in Na_2O crystal will be 6.
(C) Minimum distance between two cubical voids in simple cube unit cell lattice will be a where a is length of edge of unit cell.
(D) By defects in solids density of solids either remains constant or decreases but it can never increase.

Q20. In which of following density of lattice increases?
(A) Schottky defect
(B) Frenkel defect
(C) Impurity defect
(D) Metal excess defect

Q21. Select the correct statement(s) -
(A) The ionic crystal of AgBr has Frenkel defect
(B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90°$, $\gamma = 120°$ is hexagonal
(C) In ionic compounds having Frenkel defect the ratio r^+/r^- is low
(D) The co-ordination number of Na^+ ion in NaCl is 6

Q22. Select the correct statement (s)

(A) In NaCl crystal, each ion has an octahedral coordination.

(B) The C.N. of an atom in hcp structure is 12.

(C) Diamagnetic materials are attracted by the magnetic fields.

(D) An octahedral hole is smaller than a tetrahedral hole.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 23 to 24

The solid are characterised by incompressibility, rigidity and mechanical strength. Solids have definite volume, shape, slow diffusion, law vapour pressure and possesses the unique property of being rigid. Solids can be classified as crystalline or amorphous on basis of the nature of order present in arrangement of their constituent particles.

Q23. Which of the following is not correct for amorphous solid :-

(A) Isotropic in nature

(B) Super cooled liquid

(C) Long range order

(D) Irregular Shape

Q24. Correct option for crystalline solid is :-

(A) Isotropic Nature

(B) Rough Surface on cutting

(C) Pseudo solids

(D) Sharp M.P.

Paragraph for Question No. 25 to 26

On the basis of classification of symmetry, the lattice have been devided into seven system. These can be grouped into 7 crystal systems. With 7 crystal system, there are total 14 bravais lattice possibles.

Q25. Which of following crystal system has/have all side length equal :-

(A) Cubic (B) Tetragonal

(C) Rhombohedral (D) Triclinic

Q26. Which of following crystal system has/have body centered bravais lattice

(A) Cubic (B) Tetragonal

(C) Monoclinic (D) Orthorhombic

Paragraph for Question No. 27 to 31

In haxagonal close packing second row of spheres are placed in depressions of first row whereas third row is vertically aligned with first, 2nd with fourth forming a pattern AB. AB.....The number of nearest atoms is called the coordination number. It has six fold axis of symmetry.

Q27. What is the coordination number of a central sphere in hcp.

(A) 8 (B) 12

(C) 6 (D) 4

Q28. If $\frac{r_+}{r_-}$ lies between 0.225 to 0.414, cation occupies

(A) tetrahedral void (B) octahedral void

(C) trigonal void (D) cubic void

Q29. The coordination number of octahedral void is -

(A) 8 (B) 4

(C) 6 (D) 3

Q30. Which of the following packing is most efficient.

(A) hcp (B) fcc

(C) A&Bboth (D) bcc

Q31. ABC.ABC ..... is called

(A) hcp (B) fccp

(C) bccp (D) simple cubic

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q32. Match list-I with list-II and select the correct answer by using the codes given below :

List-I (Shape)	**List-II (Radius ratio)**
(A) Planar triangle	(I) 0.732
(B) Square planar	(II) 0.225
(C) Body centered cubic	(III) 0.155
(D) Tetrahedral	(IV) 0.414

Codes :

	A	B	C	D
(A)	III	IV	I	II
(B)	III	II	I	IV
(C)	II	I	IV	III
(D)	I	III	IV	II

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q33. Match the column :

Column-I	**Column-II**
(A) Tetragonal and Hexagonal	(P) are two crystal systems
(B) Cubic and Rhombohedral	(Q) $a = b \neq c$
(C) Monoclinic and Triclinic	(R) $a \neq b \neq c$
(D) Cubic and Hexagonal	(S) $a = b = c$

Q34. Match the crystal system / unit cells mentioned in Column I with their characteristic features metioned in Column II.

Column-I	**Column-II**
(A) simple cubic and face-centred cubic	(P) have these cell parameters $a = b = c$ and $\alpha = \beta = \gamma$
(B) cubic and rhombohedral	(Q) are two crystal systems
(C) cubic and tetragonal	(R) have only two crystallographic angles of 90°
(D) hexagonal and monoclinic	(S) belong to same crystal system

Q35. Match the crystal system/unit cells mentioned in Column I with their characteristic features mentioned in Column II.

Column I	**Column II**
(A) Simple cubic and face-centered	(P) have these cellcubic parameters
(B) cubic and rhombohedral	(Q) are two crystal systems
(C) cubic and tetragonal	(R) have only two crystallo-graphic angles of 90°
(D) hexagonal and monoclinic	(S) belong to same crystal system

Q36. Match the column :

Column I	**Column II**
[Systems]	**[Packing fraction]**
(A) Simple cubic	(P) 0.74
(B) FCC	(Q) 0.68
(C) BCC	(R) 0.414
(D) hcp	(S) 0.52

Q37. Match the column :

Column I	**Column II**
(A) Rock salt structure	(P) Co-ordination number of cation is 4
(B) Zinc Blend structure	(Q) $\frac{\sqrt{3}a}{4} = r_+ + r_-$
(C) Flourite structure	(R) Co-ordination number of cation and anion are same
	(S) Distance between two nearest anion is $\frac{a}{\sqrt{2}}$

Q38. Match the column :

Column I	**Column II**
(A) If radius ratio $x\left(\frac{r_c}{r_a}\right) < 0.155$	(P) Coordination no. is 8
(B) If $0.225 \le x < 0.414$	(Q) Coordination no. is 4
(C) If $0.414 \le x < 0.732$	(R) Coordination no. is 6
(D) If $0.732 \le x < 1$	(S) Coordination no. is 2

Q39.

Column-I	**Column-II**
(A) H_2O(ice)	(P) Insulator
(B) Fe	(Q) Molecular Solid
(C) $C_{(graphite)}$	(R) Conductor
(D) ZnS (Solid)	(S) Metallic Solid
	(T) Covalent Solid

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q40. **Statement 1 :** Glass is isotropic
Statement 2 : Glass is supercooled liquid.

Q41. **Statement 1 :** Diamond is a covalent solid.
Statement 2 : The coordination number of each carbon atom in diamond is 4.

Q42. **Statement 1 :** Stoichiometric compounds obey the law of constant composition.
Statement 2 : Schottky and Frenkel defects are observed in stoichiometric compounds.

Q43. **Statement 1 :** An ionic structure is composed of oppositely charged ions.
Statement 2 : If the larger ions are close packed then the smaller ions may occupy either the octahedral holes or the tetrahedral holes depending on their size.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

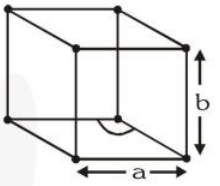
Q44. Number of solids, that are amorphous from given below polyurethane, Teflon, Potassium nitrate, Polyvinyle Chloride, Fibre glass, Copper.

Q45. Total number of bravais lattice of cubic crystal system is -

Q46. If C.N. in S.C. is x, C.N. is BCC is y and C.N. in FCC is Z then value of $\frac{x \times y}{z}$ is -

Q47. A solid crystallises in cubic crystal in which 'X' atom occupy all the corners and body centres and y atoms occupy all the face - centres and simplest formula of solid is X_aY_b then find value of (a + b).

Q48. Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined, the lattice constants were a = 4.53Å, and b = 7.60Å (see figure). How many molecules are contained in a given unit cell? [density (ice) = 0.92 gm/cm^3]



Q49. An element crystallizes in BCC structure. The edge of its unit-cell is 288 pm. If the density of the crystals is 7.2 g cm^{-3}, what is the atomic mass of the element (in gm/mole)?

Q50. The edge length of unit cell of a metal having molecular weight 75 g/mol is 5Å which crystallizes in cubic lattice. If the density is 2 gm/cm^3 then find the radius of metal atom. ($N_A = 6 \times 10^{23}$).
Give the answer in pm.

Q51. One unit cell of a mineral is composed of A^{+2} and C^{-2} and B^{+n}. A^{+2} ions at corners, C^{-2} ions at face centres and B^{+n} at the centre of the cube are present. What is the oxidation number (+n) of B^{+n} ion.

Q52. Calculate the number of atoms in a cubic based unit-cell having one atom on each corner and two atoms one each body diagonal.

Q53. Find the percentage void in HCP unit cell.

Q54. If the radius of Mg^{2+} ion, Cs^+ ion, O^{2-} ion, S^{2-} ion and Cl^- ion are 0.65 Å, 1.69 Å, 1.40 Å, 1.84 Å, and 1.81Å respectively. Calculate the sum of co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl.

Q55. Number of Na^+ present in per unit cell of Na_2O is?

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Amorphous substances show :
(1) short and long range order
(2) short range order
(3) long range order
(4) have no sharp m.p.
(A) 1 and 3 are correct
(B) 2 and 3 are correct
(C) 3 and 4 are correct
(D) 2 and 4 are correct

Q2. How many unit cells are present in 39 gm of potassium that crystallies in body centred cubic structure? [At.wt. of K = 39]

(A) 6.02×10^{23} (B) $\frac{6.02\times10^{23}}{4}$

(C) $\frac{6.02\times10^{23}}{2}$ (D) $\frac{3}{4}\times6.02\times10^{23}$

Q3. The crystal system of a compound with unit cell dimensions, a = 0.387, b = 0.387 and c = 0.504 nm, and $\alpha = \beta = 90°$ and $\gamma = 120°$ is :-
(A) Cubic
(B) Hexagonal
(C) Orthorhombic
(D) Rhombohedral

Q4. Which of the following are the correct axial distances and axial angles for rhombohedral system?
(A) $a = b = c$, $\alpha = \beta = \gamma \neq 90°$
(B) $a = b \neq c$, $\alpha = \beta = \gamma = 90°$
(C) $a \neq b \neq c$, $\alpha = \beta = \gamma = 90°$
(D) $a \neq b \neq c$, $\alpha \neq \beta \neq \gamma \neq 90°$

Q5. An oxide of 'A' crystallizes with eight atoms at the corners of the unit cell and 18 oxygen atoms on the edges & on the faces symmetrically. What is the formula of this oxide -
(A) AO_6 (B) A_2O_4
(C) A_3O_6 (D) A_3O_8

Q6. A solid compound contains X, Y and Z atoms in a cubic lattice with X atoms occupying the corners, Y atoms in the body centred positions and Z atoms at the centre of the faces of the unit cell. What is the empirical formula of the compound ?
(A) XY_2Z_3 (B) XYZ_3
(C) $X_2Y_2Z_3$ (D) X_8YZ_6

Q7. An alloy of metal 'A' 'B' and 'C' is found to have 'A' constituting ccp lattice. If 'B' atom occupy the edge centres and 'C' is present at the body centre, the formula of the alloy is :-
(A) A_4B_2C (B) A_4B_4C
(C) A_4B_3C (D) ABC

Q8. Packing fraction in 2-D hexagonal arrangement is -

(A) $\frac{\pi}{3\sqrt{2}}$ (B) $\frac{\pi}{3\sqrt{3}}$

(C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi}{6}$

Q9. For an ionic crystal of the general formula AX and coordination number 6, the values of radius ratio will be?
(A) > .73 (B) 0.41 < _ < 0.73
(C) 0.22 < _ < 0.41 (D) < 0.22

Q10. If the ratio of co-ordination number of A that of B is X : Y, then the ratio of number of atoms of A to that number of atoms of B in unit cell is -
(A) X : Y (B) Y : X
(C) X^2 : Y (D) Y : X^2

Q11. An element cystallizes in both fcc & bcc lattice. If the density of the element in the two forms is same, the ratio of unit cell edge length of fcc to that of bcc lattice is :-

(A) $(2)^{1/3}$ (B) $(1/2)^{1/3}$

(C) $(4)^{1/3}$ (D) $(1/4)^{1/3}$

Q12. When heated above 916°C, iron changes its bcc crystalline form to fcc without the change in the radius of atom. The ratio of density of the crystal before heating and after heating is :-

(A) 1.069 (B) 0.918

(C) 0.725 (D) 1.231

Q13. Sodium metal exists in bcc unit cell. The distance between nearest sodium atoms is 0.368 nm. The edge length of the unit cell is:

(A) 0.368 nm (B) 0.184 nm

(C) 0.575 nm (D) 0.425 nm

Q14. First three nearest neighbour distances for body centered cubic lattice are respectively (edge length of unit cell = a) :-

(A) $\sqrt{2}a, a\sqrt{3}a$ (B) $\frac{a}{\sqrt{2}}, a, \sqrt{3}a$

(C) $\frac{\sqrt{3}a}{2}, a, \sqrt{2}a$ (D) $\frac{\sqrt{3}a}{2}, a, \sqrt{3}a$

Q15. The number of nearest neighbours around each particle in a face-centred cubic lattice is :-

(A) 4 (B) 6

(C) 8 (D) 12

Q16. Platinum crystallizes in a face-centered cubic crystal with a unit cell edge length 'a'. The distance between nearest neighbour atom is :

(A) a (B) $a\frac{\sqrt{3}}{2}$

(C) $a\frac{\sqrt{2}}{2}$ (D) $a\frac{\sqrt{2}}{4}$

Q17. A crystal is made of particles X, Y and Z forms fcc packing. Y occupies all the octahedral voids of X and Z occupies all the tetrahedral voids of X. If all the partical along one body diagonal are removed then the formula of the crystal would be :-

(A) XYZ_2 (B) X_2YZ_2

(C) $X_8Y_4Z_5$ (D) $X_5Y_4Z_8$

Q18. A compound is formed by three element A, B and C. The element A form ccp, atoms of B occupy 2/3rd of tetrahedral void and atoms of c occupy 1/3rd of octrahedral voids then formula of the compound is :-

(A) A B C (B) $A_3B_4C_3$

(C) A_3B_4C (D) A_4B_3C

Q19. In a compound, atoms of element Y form ccp lattice and those of element X occupy 2/3rd of tetrahedral voids. The formula of the compound will be :-

(A) X_4Y_3 (B) X_2Y_3

(C) X_2Y (D) X_3Y_4

Q20. In a structure of mixed oxide, oxide ions are in CCP. One fifth of tetrahedral voids are occupied by divalent ion (A^{2+}) while half of the octahedral voids are occupied by trivalent ion (B^{3+}), the formula is :-

(A) $A_4B_5O_{10}$ (B) A_2BO_4

(C) AB_2O_4 (D) $A_5B_4O_{10}$

Q21. A certain oxide of metal M crystallises in such a way that O^{-2} ions occupy hcp arrangement following ABAB----------pattern. The metal ions however occupy 2/3rd of the octahedral voids then formula of the compound is –

(A) M_2O_3 (B) M_3O

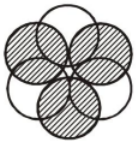
(C) MO_2 (D) M_3O_2

Q22. Octahedral void is surrounded by .....spheres :

(A) 6 (B) 4

(C) 8 (D) 12

Q23. The empty space between the shaded balls and hollow balls as shown in the diagram is called :-



(A) Hexagonal void
(B) octahedral void
(C) Tetrahedral void
(D) Double triangular void

Q24. A binary solid (A^+B^-) has a rock salt structure. If the edge length is 400 pm and radius of cation is 75 pm, the radius of anion is :-

(A) 100 pm (B) 125 pm
(C) 250 pm (D) 325 pm

Q25. KCl crystallizes in the same type of lattice as does NaCl. Given that $r_{Na^+} / r_{Cl^-} = 0.50$ and $r_{Na^+} / r_{K+} = 0.70$, calculate the ratio of the edge length of unit cell for KCl to that for NaCl :-

(A) 1.143 (B) 1.224
(C) 1.414 (D) None

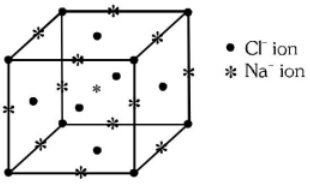
Q26. If the radius ratio is in the range of 0.414 - 0.732 then the co-ordination number will be :

(A) 2 (B) 4
(C) 6 (D) 8

Q27. The two ions A^+ and B^- have radii 88 and 200 pm respectively. In the closed packed crystal of compound AB, predict the co-ordination number of A^+.

(A) 6 (B) 8
(C) 4 (D) None

Q28.



In this NaCl type unit cell anion Cl^- adopts :-

(A) B.C.C.
(B) AB AB AB ..... type
(C) ABC ABC ABC .....type
(D) ABC CBA ABC CBA .....type

Q29. CaO and NaCl have the same crystal structure and nearly the same ionic radii. If x is the lattice energy of NaCl, the lattice energy of CaO is very nearly :-

(A) 2x (B) x
(C) 4x (D) $\frac{x}{2}$

Q30. In a compound XY the radius of cation X^+ and Y^- are 1.69Å and 1.81Å respectively. What will be the ratio of coordination numbers for X^+ and Y^-.

(A) 6 : 6 (B) 1 : 1
(C) 4 : 4 (D) 8 : 8

Q31. What type of crystal defect is indicated in the diagram below ?

Na^+ Cl^- Na^+ Cl^- Na^+ Cl^-
Cl^- □ Cl^- Na^+ □ Na^+
Na^+ Cl^- □ Cl^- Na^+ Cl^-
Cl^- Na^+ Cl^- Na^+ □ Na^+

(A) Frenkel defect
(B) Schottky defect
(C) Interstitial defect
(D) Frenkel and Schottky defects

Q32. NaCl is doped with 2×10^{-3} mol% $SrCl_2$. The concentration of cation vacancies is :

[Hint : Concentration of vacancies = $\frac{2\times10^{-3}}{100} N_A$]

(A) 3.01×10^{18} mol^{-1}
(B) 12.04×10^{18} mol^{-1}
(C) 6.02×10^{18} mol^{-1}
(D) 12.04×10^{20} mol^{-1}

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statement(s) is/are incorrect for NaCl lattice ?

(A) An Na^+ ion is placed at a distance (1/6)a directly above each Cl^- ion (where a is a cubic unit-cell edge length)

(B) An Na^+ ion is placed at a distance (1/4)a directly above each Cl^- ion

(C) An Na^+ ion is placed at a distance of $\frac{1}{2\sqrt{2}}$ a directly above each Cl^- ion

(D) An Na^+ ion is placed at a distance $\frac{1}{\sqrt{2}}$ a directly above each Cl^- ion

Q2. A cubic solid is made up of two elements P and Q. Atoms Q are present at the corners of the cube and atoms of P at the body centre. Which of the following is/are correct ?

(A) The formula of the compound is PQ

(B) The coordination number of P and Q are the same

(C) Coordination number of Q = 6

(D) Coordination number of P = 8

Q3. Which of the following statement(s) is/are correct –

(A) NaCl is a 'AB' crystal lattice that can be interpreted to be made up of two individual fcc unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.

(B) In a face centred cubic unit cell, the body centre is an octahedral void.

(C) In fcc unit cell, octahedral and tetrahedral voids are equal in number.

(D) Tetrahedral voids = 2 × octahedral voids, is valid for ccp and hcp.

Q4. Select the correct statement(s) with respect to Rock-Salt structure of ionic compounds :

(A) Co-ordination number of cation is four whereas that of anion is six.

(B) Co-ordination number of cation is six whereas that of anion is four.

(C) Co-ordination number of each cation & anion is four.

(D) Co-ordination number of each cation & anion is six.

Q5. In an AB unit crystal of NaCl types assuming Na^+ forming FCC

(A) The nearest neighbour of A^+ is $6B^-$ ion

(B) The nearest neighbour of B^- is 6 A^+ ion

(C) The second neighbour of A^+ is 12 A^+ ion

(D) The packing fraction of AB crystal is $\frac{\sqrt{3}\pi}{8}$

Q6. Which of the following statement (s) is/are correct?

(A) The coordination number of each type of ion in CsCl crystal is 8.

(B) A metal that crystallizes in bcc structure has a coordination no. of 12.

(C) A unit cell of an ionic crystal shares some of its ions with other unit cells.

(D) The length of the unit cell in NaCl is 552 pm (r_{Na^+} = 95pm, r_{Cl^-} = 181pm)

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 7 to 9

Density of a unit cell is represented as

$$\rho = \frac{\text{Effective no. of atoms (s)} \times \text{Mass of a unit cell}}{\text{Volume of a unit cell}}$$

$$= \frac{Z.M.}{N_A.a^3}$$

where, mass of unit cell = mass of effective no. of atoms (s) or ion (s).

M = At. wt./formula wt.

N_A = Avogadro's no. $\Rightarrow 6.023 \times 10^{23}$

a = edge length of unit cell

Q7. Silver crystallizes in a fcc lattice and has a density of 10.6 g/cm^3. What is the length of an edge of the unit cell ?

(A) 0.407 nm (B) 0.2035 nm

(C) 0.101 nm (D) 4.07 nm

Q8. An element crystallizes in a structure having fcc unit cell of an edge 200 pm. Calculate the density, if 100 g of this element contains 12×10^{23} atoms :

(A) 41.66 g/cm^3 (B) 4.166 g/cm^3

(C) 10.25 g/cm^3 (D) 1.025 g/cm^3

Q9. The density of KBr is 2.75 g/cm^{-3}. The length of the edge of the unit cell is 654 pm. To which type of cubic crystal, KBr belongs ?

(A) simple cubic (B) bcc

(C) fcc (D) none of these

Paragraph for Question No. 10 to 13

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (hcp), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assumed radius of every sphere to be 'r'.

Q10. The number of atoms in this hcp unit cell is -

(A) 4 (B) 6 (C) 12 (D) 17

Q11. The volume of this hcp unit cell is -

(A) $24\sqrt{2}r^3$ (B) $16\sqrt{2}r^3$

(C) $12\sqrt{2}r^3$ (D) $64r^3 / 3\sqrt{3}$

Q12. The empty space in this hcp unit cell is -

(A) 74% (B) 47.6%

(C) 32% (D) 26%

Q13. The arrangement in hcp unit cell is of which type?

(A) AAA.... (B) ABABAB....

(C) ABCABC.... (D) ABBAABBA...

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14. Match the column :

Column-I **[Bravais lattice(s)]**	**Column-II** **[Crystal system]**
(A) Primitive, face centered, body centered, end centered	(P) Cubic
(B) Primitive, face centered, body centered	(Q) Orthorhombic
(C) Primitive, body centered	(R) Hexagonal
(D) Primitive only	(S) Tetragonal

Q15.

Column-I	**Column-II**
(A) S.C	(P) % packing efficiency = 74%
(B) B.C.C	(Q) % packing efficiency = 52%
(C) F.C.C	(R) Coordination number = 8
(D) Monoclinic	(S) At least one angle ≠ 90°

Q16.

Column-I (Ionic Crystal)	**Column-II**
(A) CsCl	(P) $r + r^- = \frac{a\sqrt{3}}{4}$
(B) CaF_2	(Q) C.N (cation) = 4
(C) ZnS	(R) C.N (anion) = 8
(D) Na_2O	(S) C.N (anion) = 4

Q17. Match the column :

Column-I	**Column-II**
(A) Simple vacancy defect	(P) Shown by non-ionic solids and increases density of the solide
(B) Simple interstitial defect	(Q) Shown by ionic solids and decreases density of the solid
(C) Frenkel defect	(R) Shown by ionic solids and density of the solid decreses
(D) Schottky defect	(S) Shown by ionic solids and density of the solid remains the same

TRIPLE MATCH TYPE

*This section contains TWO tables (each having 3 columns and 4 rows). Based on each table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column - I	**Column - II**	**Column - III**
(P) Cubic	(I) a = b = c	(A) No. of Bravais lattice = 4
(Q) Hexagonal	(II) $\alpha = \beta = \gamma$	(B) No. of Bravais lattice = 1
(R) Rhombohedral	(III) $a \neq b \neq c$	(C) Body centered
(S) Orthorhombic	(IV) $\alpha \neq \beta \neq \gamma$	(D) Primitive

Q18. Incorrect combination is -

(A) P - I - D (B) P - II - C (C) R - II - B (D) S - II - B

Q19. Choose correct combination out of these :-

(A) P - I - A (B) R - I - B (C) S - III - B (D) Q - IV - B

Q20. In which of the following is correct for all 3 Column :-

(A) Q - II - B (B) Q - III - D (C) P - II - D (D) R - II - C

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q21. **Statement 1 :** ZnO becomes yellow when it is heated.

Statement 2 : NaCl becomes yellow when heated in the presence of Na vapours due to anion vacancy.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q22. Find the number of tetrahedral voids in hexagonal primitive unit cell.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	D	A	A	A	C	A	A	C	D
11	12	13	14	15	16	17	18	19	20
D	A	C	C	D	C	B	B	C	A
21	22	23	24	25	26	27	28	29	30
B	B	A	A	A	B	A	B	B	B
31	32	33	34	35	36	37	38	39	40
B	C	A	C	C	C	A	D	A	A
41	42	43	44	45	46	47	48	49	50
B	C	A	C	B	B	A	D	D	C
51	52	53	54	55	56	57	58	59	60
B	B	C	B	A	B	A	D	D	C
61	62	63	64						
D	B	D	B						

Ex-1A

1. C, D **2.** A, B, C **3.** A, C **4.** B, D **5.** C
6. A **7.** B, C **8.** A, B, D **9.** D **10.** A, B, D
11. A, B, C, D **12.** A, B, D **13.** B **14.** A, B, C, D **15.** A, B, C, D
16. A, B **17.** C, D **18.** A **19.** B, C **20.** C, D
21. B, D **22.** A, B **23.** C **24.** D **25.** A, C
26. A, B, D **27.** B **28.** A **29.** C **30.** C
31. B **32.** A
33. (A) → P, Q ; (B) → P,S ; (C) → P,R ; (D) → P
34. (A) P, S ; (B) -P,Q ; (C) - Q ; (D) - Q,R
35. (A) → P,S ; (B) → P,Q ; (C)→ Q ;(D)→ Q,R
36. (A) → S ; (B) → P ; (C)→ Q ;(D)→ P
37. (A) → R,S ; (B) → P,Q,R,S ; (C) → Q
38. (A) → S ; (B) → Q ; (C)→ R ; (D)→ P
39. (A) P, Q; (B) R, S; (C) R, T; (D) P
40. A **41.** B **42.** B **43.** B **44.** 4
45. 3 **46.** 4 **47.** 5 **48.** 4 **49.** 52
50. 2.16Å **51.** 4 **52.** 9 **53.** 26 **54.** 18
55. 8

Ex-2

1	2	3	4	5	6	7	8	9	10
D	C	B	A	A	B	C	C	B	B
11	12	13	14	15	16	17	18	19	20
A	B	D	C	D	C	D	C	A	A
21	22	23	24	25	26	27	28	29	30
A	A	B	B	A	C	A	C	C	D
31	32								
B	B								

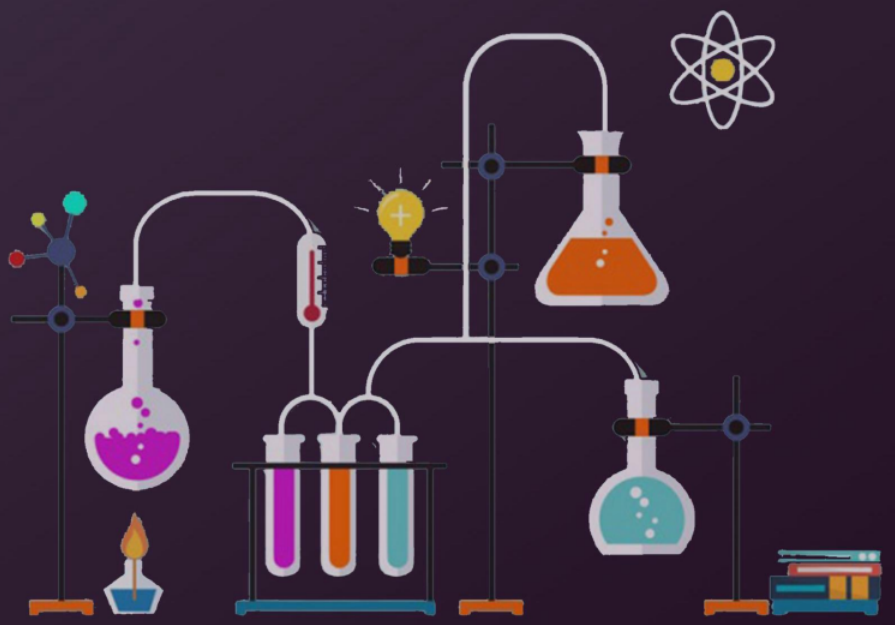
Ex-2A

1. A, C, D **2.** A, B, D **3.** A, B, D **4.** D **5.** A, B, C
6. A, C, D **7.** A **8.** A **9.** C **10.** B
11. A **12.** D **13.** B
14. (A) → Q; (B) → P; (C) → S; (D) → R
15. (A) → Q; (B) → R; (C) → P; (D) → S
16. (A) - R; (B) - P,S; (C) - P,Q,S; (D) - P, Q, R
17. (A) → R ; (B) → P ; (C) → S ; (D) → Q
18. D **19.** B **20.** C **21.** B **22.** 12

-: Important Notes & Formulas :-

CHEMISTRY

Solid State



SHEET

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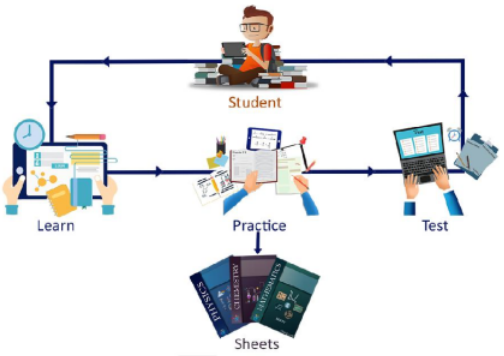
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction of Solid State	1			
2	Types of Solid	2-5	23,24,44,40	1	
3	Lattice and Unit Cells	6,7		2	
4	Bravais Lattices	8-10	1,2,25,26, 33-35,45	3,4	14,18-20
5	Coordination number and Packing Fraction	11-17	36, 46, 47,41	5-10	1,2,15
6	Packing in FCC and Density of Crystals	18-26	7,48-50	11,12	7-9
7	Close Packing in 1D, 2D and Square Close Packing in 3D	27-29	6,8-11	13-16	10-12
8	Hcp, Ccp and Body centered Packing	30-35	3-5, 27-31,51-53		13
9	Octahedral and Tetrahedral Voids	36-42	12,13	17-23	22
10	Packing in Ionic Solids	43-46	14,32,54	24-27	
11	Structures of Ionic Solids	47-55	15-18,37, 38,55	28-30	3-6,16,17
12	Imperfections in Solids, Frenkel and Schottky Defects	56-63	19-21,42,43	31	21
13	Electrical Properties of Solids		22,39		
14	Magnetic Properties of Solids	64			
15	ACC			32	

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Solid State

Q1. Which of the following is not the characteristic of solid :-
(A) fixed shape
(B) fixed volume
(C) low thermal energy
(D) low density

Types of Solid

Q2. A pseudo solid is :
(A) glass
(B) pitch
(C) KCl
(D) Glass and pitch both

Q3. Crystalline solids are generally in nature :-
(A) Rigid (B) Flexible
(C) Soft (D) None of these

Q4. Crystalline solids are?
(A) Anisotropic (B) Isotropic
(C) Both A and B (D) None of these

Q5. Amorphous solids may be classified as :-
(A) isotropic and super-cooled liquids
(B) anisotropic and super-cooled liquids
(C) isoenthalpic and super-heated liquids
(D) isotropic and super-heated solids.

Lattice and Unit Cell

Q6. The three dimensional graph of lattice points which sets the pattern for the whole lattice is called:-
(A) Space lattice (B) Simple lattice
(C) Unit cell (D) Crystal lattice

Q7. The number of atoms per unit cell in a simple cubic, face-centred cubic and body-centered cubic structure respectively are -
(A) 1, 4, 2 (B) 1, 2, 4
(C) 8, 14, 9 (D) 8, 4, 2

Bravais Lattices

Q8. Which of the following are the correct axial distances and axial angles for rhombohedral system?
(A) $a = b = c, \alpha = \beta = \gamma \neq 90°$
(B) $a = b \neq c, \alpha = \beta = \gamma = 90°$
(C) $a \neq b \neq c, \alpha = \beta = \gamma = 90°$
(D) $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90°$

Q9. Which of the following crystal system has more than one bravais lattice -
(A) Lattice with unit cell triclinic
(B) Lattice with unit cell hexagonal
(C) Lattice with unit cell monoclinic
(D) Lattice with unit cell rhombohedral

Q10. $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90°$ represents.
(A) tetragonal system
(B) orthorhombic system
(C) monoclinic system
(D) triclinic system

Coordination Number And Packing Fraction

Q11. In a face centred cubic cell, an atom at the corner contributes to the unit cell,
(A) 1 part (B) 1/2 part
(C) 1/4 part (D) 1/8 part

Q12. The coordination number of hexagonal closest packed(hcp) structure is
(A) 12 (B) 10
(C) 8 (D) 6

Q13. The space occupied in bcc arrangement is -
(A) 74% (B) 70%
(C) 68% (D) 60.4%

Q14. Lithium crystallizes as body centered cubic crystals. If the length of the side of unit cell is 350 pm, the atomic radius of lithium is :-
(A) 303.1 pm (B) 606.2 pm
(C) 151.5 pm (D) 123.7 pm

Q15. Body centred cubic and face centred cubic unit cells have n_1 and n_2 effective number of atoms respectively. Which one of the following $[n_1, n_2]$ combination is correct:
(A) [4, 1] (B) [4, 2]
(C) [1, 4] (D) [2, 4]

Q16. Which one of the following has highest packing efficiency ?
(A) Simple cubic
(B) Body centered cubic
(C) Hexagonal closed packing
(D) All have equal

Q17. $d = \frac{Z \times M}{N_A . a^3}$ $d \alpha z$ (Keeping other parameter fix)
Simple cubic Z = 1; B.C.C Z = 2
Hexagonal close packing Z = 6
Al (at wt. 27) crystallizes in the body centred cubic system with a = 4.05 Å.
Calculate the diameter of Al atom :-
(A) 1.732Å (B) 3.507Å
(C) 0.825Å (D) 1.650Å

Packing in FCC and Density of Crystals

Q18. An element (atomic mass = 60) having face centred cubic crystal has a density of 6.23 g cm^{-3}. What is the edge length of the unit cell
(Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1})
(A) 3×10^{-8} (B) 4×10^{-8}
(C) 6×10^{-8} (D) 2×10^{-8}

Q19. In a face centred cubic arrangement of A and B atoms where A are present at the corner and B at the face centres, A atoms are missing from 4 corners in each unit cell. What is the simplest formula of the compound ?
(A) AB_3 (B) A_2B_2 (C) AB_6 (D) AB_4

Q20. An element crystallises into a structure which may be described by a cubic type of a unit cell having one atom on each corner of the cube and two atoms on one of its diagonals. If the volume of this unit cell is 24×10^{-24} cm^3 and the density of the element is 7.2 g cm^{-3}, calculate the number of atoms present in 200g of the element ?
(A) 3.472×10^{24} (B) 1.1513×10^{24}
(C) 10.46×10^{24} (D) 1.8069×10^{24}

Q21. The face diagonal of a cubic closed packed unit cell is 4 Å. What is the length?
(A) 2.85 Å (B) 2.83 Å
(C) 2.86 Å (D) 2.87 Å

Q22. Silver crystallises in FCC lattice. If edge length of the cell is 4.077×10^{-8} cm and density is 10.5 g cm^{-3}, calculate the atomic mass of silver.
(A) 107.18 (B) 107.14
(C) 107.10 (D) 107.17

Q23. An element crystallises in BCC structure. The edge length of its unit cell is 288 pm. If the density of the crystal is 7.2 g cm^{-3}, what is the atomic mass of the element ?
(A) 51.8 (B) 103.6 (C) 25.9 (D) 207.2

Q24. The available space occupied by spheres of equal size in three dimensions in both hcp and ccp arrangement is
(A) 74% (B) 70% (C) 60.4% (D) 52.4%

Q25. A compound formed by elements A and B crystallises in cubic structure, in which atoms of A are at the corners while that of B are at the face centre. The formula of the compound is?
(A) AB_3 (B) AB
(C) A_3B (D) none of these

Q26. Each edge of a cubic unit cell is 400 pm long. If atomic weight of the element is 120 and it's density is 6.25 g/cm^3. The crystal lattice is? ($N_A = 6\times10^{23}$)
(A) Primitive cell (B) body centered
(C) face centered (D) All of these

Close Packing In 1D, 2D and Square Close Packing in 3D

Q27. What are the number of atoms per unit cell and the number of nearst neighbours in a simple cubic structure :-
(A) 1, 6 (B) 4, 12 (C) 2, 8 (D) 2, 6

Q28. If 5 Rs. coins are aaranged in square 2D packing then the packing efficiency is -
(A) 22% (B) 78.54%
(C) 52% (D) 68%

Q29. If identical spheres are arranged, in 2D in such a way bottom row sphere fits in dipression of top row, then the packing efficiency and co-ordination number is :
(A) 60%, 6 (B) 90%, 6
(C) 60%, 4 (D) 90%, 4

HCP ,CCP and Body Centered Packing

Q30. If the radius of an atom of an element is 75 pm and the lattice is body-centred cubic, the edge of the unit cell will be?
(A) 32.475 pm (B) 173.41 pm
(C) 37.5 pm (D) 212.1 pm

Q31. In a body centred unit cell, one of the diagonals contains two extra atoms. Find the number of atoms present per unit cell ?
(A) 1 (B) 4 (C) 2 (D) 6

Q32. The hcp and ccp structure for a given element would be expected to have :-
(A) the same alignment of layers
(B) the same density
(C) the same packing fraction
(D) all the above

Q33. Iron (At. mass = 56) has the bcc crystal lattice structure. Edge length of its unit cell is 278 pm. Radius of iron atom would be :-
(A) 120 pm (B) 278 pm
(C) 150 pm (D) 210 pm

Q34. Which of the following layering pattern will have a void fraction of 0.260 ?
(A) ABCCBAABC (B) ABBAABBA
(C) ABCABCABC (D) ABCAABCA

Q35. In the body-centered cubic unit cell & face centered cubic unit cell, the radius of atom in terms of edge lengh(a) of the unit cell is respectively :
(A) $\frac{a}{2}, \frac{a}{2\sqrt{2}}$ (B) $\frac{a}{2\sqrt{2}}, \frac{\sqrt{3}a}{4}$
(C) $\frac{\sqrt{3}a}{4}, \frac{a}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}a}{2}, \frac{a}{2\sqrt{2}}$

Octahedral and Tetrahedal Voids

Q36. The number of octahedral sites in a cubical close packed array of N spheres is?
(A) N/2 (B) 2N
(C) N (D) 4N

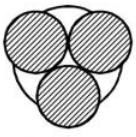
Q37. Atom A is every element of FCC, atom B is present at every Octahedral void, atom C is present at 25% of Tetrahedral void. Find out the possible molecular formula of the compound ?
(A) A_2B_2C (B) A_3B_3C
(C) A_4B_4C (D) A_5B_5C

Q38. The co-oridnation number of a cation occupying an octahedral hole is?
(A) 4 (B) 6 (C) 8 (D) 12

Q39. In a crystalline solid anion B are arranged in ccp lattice and cation A occupy 50% of the octahedral void and 50% of the tetrahedral voids. What is the formula of the solid ?
(A) A_3B_2 (B) AB
(C) A_2B_3 (D) A_4B_3

Q40. In a multi layered close-packed structure :
(A) there are twice as many tetrahedral holes as there are close-packed atoms
(B) there are as many tetrahedral holes as there are closed packed atoms
(C) there are twice as many octahedral holes as there are close-packed atoms
(D) there are as many tetrahedral holes as there are octahedral holes

Q41. The empty space between the shaded balls and hollow balls as shown in the diagram is called :-



(A) Hexagonal void
(B) octahedral void
(C) Tetrahedral void
(D) Double triangular void

Q42. If the anion (A) form hexagonal closest packing and cation (C) occupy only 2/3 octahedral voids in it, then the general formula of the compound is :-
(A) CA (B) CA_2
(C) C_2A_3 (D) C_3A_2

Packing in Ionic Solids

Q43. A solid A^+B^- has NaCl type close packed structure. If the anion has a radius of r_a 250 pm, what should be the ideal radius for the cation ? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal A^+B^- ?
(A) 56.25pm (B) 55.26pm
(C) 57.20pm (D) 56.20pm

Q44. If the radius of an octahedral void is r and the radius of the atom in close packing is R, derive the relation between r and R ?
(A) r = 0.412 R (B) r = 0.410 R
(C) r = 0.414 R (D) r = 0.418 R

Q45. In NaCl type structure shortest distance between two nearest neighbours is 100 pm, then distance between two next nearest neighbours in the same unit cell will be :
(A) $\frac{100}{\sqrt{2}}$pm (B) $100\sqrt{2}$pm
(C) 200 pm (D) $200\sqrt{2}$pm

Q46. The limiting radius ratio for tetrahedral shape is :-
(A) 0.414 to 0.732 (B) 0.225 to 0.414
(C) 0.155 to 0.225 (D) 0 to 0.155

Structures of Ionic Solids

Q47. Diamond belongs to the cyrstal system :
(A) Cubic (B) triclinic
(C) tetragonal (D) hexagonal

Q48. Which one of the following statements is incorrect about rock salt type :-
(A) It has FCC arrangement of Cl
(B) Na^+ and Cl^- ions have a coordination number of 6:6
(C) A unit cell of NaCl consists of four NaCl units
(D) All halides of alkali metals have rock salts type structure

Q49. The tetrahedral voids formed by ccp arrangement of Cl^- ions in rock salt structure are :
(A) Occupied by Na^+ ions
(B) Occupied by Cl^- ions
(C) Occupied by either Na^+ or Cl^- ions
(D) Vacant

Q50. The co-ordination number of Cs^+ and Cl^- ions in CsCl structure is :
(A) 4 : 4 (B) 6 : 6
(C) 8 : 8 (D) 4 : 8

Q51. Select the correct statement(s):-
(a) Co-ordination number of Cs^+ and Cl^- are 8 : 8 in CsCl crystal.
(b) For fcc lattice, radius of metal atom $= \frac{a}{2\sqrt{2}}$
(c) For triclinic system $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90°$
(d) In a fcc unit cell the body centre is an octrahedral vioid
Correct code is :-
(A) a, b, c (B) a, b, c, d
(C) b, c, d (D) a, c, d

Q52. Which one is the incorrect relation :-
(A) Number of tetrahedral voids = 2 × number of atoms present in closed placed arrengement
(B) Number of tetrahedral voids = number of octahedral voids
(C) Co-ordination number of Cs^+ in CsCl is 8
(D) Radius of octahedral voids = 0.414R

Q53. In the fluorite structure, the coordination number of Ca^{2+} ion is :-
(A) 4 (B) 6
(C) 8 (D) 3

Q54. Which of the following expressions is correct in case of a sodium chloride unit cell (edge lenght, a) ?
(A) $r_c + r_a = a$ (B) $r_c + r_a = a/2$
(C) $r_c + r_a = 2a$ (D) $r_c + r_a = \sqrt{2}a$

Q55. The unit cell of Caesium Chloride has one Cs^+ ion and
(A) One Cl^- (B) Two Cl^-
(C) Three Cl^- (D) Four Cl^-

Imperfections In Solids, Frenkel and Schottky Defects

Q56. In the Schottky defect
(A) cations are missing from the lattice sites and occupy the interstitial sites
(B) equal number of cations and anions are missing
(C) anions are missing and electrons are present in their place
(D) equal number of extra cations and electrons are present in the interstitial sites

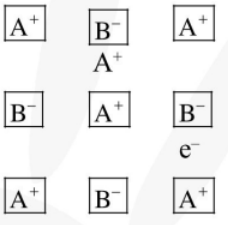
Q57. Ionic solids with Schottky defects contain in their structure,
(A) Equal number of cation and anion vacancies
(B) Anion vacancies and interstitial anions
(C) cation vacancies
(D) Cation vacancies and interstitial cations

Q58. At zero kelvin, most of the ionic crystals posses,
(A) Frenkel defect
(B) Schottky defect
(C) Metal excess deffect
(D) No defect

Q59. Frenkel defect is generally observed in,
(A) AgBr (B) Agl
(C) ZnS (D) All of these

Q60. In an ionic solid lattice,the cation has left a lattice site and is located at interstitial position, the lattice defect is :
(A) interstitial defect (B) vacancy defect
(C) Frenkel defect (D) Schottky defect

Q61. What type of crystal defect in indicated in the diagram below :-



(A) Frenkel defect (B) Schottky defect
(C) Both of them (D) None of these

Q62. Schottky defect causes :-
(A) Increase in the density of solid
(B) Decrease in the density of solid
(C) No change in the density of solid
(D) Decrease in the conductivity of solids

Q63. Select the incorrect statement -
(A) F-centres are a factor for imparting the colour to the crystal.
(B) Stoichiometry of crystal remains uneffected due to schottky defect.
(C) In orthorhombic crystal each interfacial angle is 90°.
(D) There are four unit cells in tetragonal crystal system.

Magnetic Properties of Solids

Q64. Anti ferromagnetism is present in :-
(A) ↑↑↑↑↑ (B) ↑↓↑↓
(C) Both the above (D) None

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following have two Bravais Lattice ?
(A) Cubic (B) Orthorhombic
(C) Tetragonal (D) Monoclinic

Q2. Which type of crystals contain one Bravais lattice ?
(A) hexagonal (B) triclinic
(C) rhombohedral (D) monoclinic

Q3. In a body centred unit lattice of A_2 type (Given : R = radius of atom A)
(A) the edge length is equal to $\frac{4R}{\sqrt{3}}$
(B) The edge length is equal to $2R^2 + 2R^2$
(C) The edge length is equal to $\frac{2d}{\sqrt{3}}$, where d = nearest neighbour distance
(D) The square of the edge length is equal to $16R^2$ where R is the radius of atom

Q4. The volume occupied by the atoms in F.C.C. unit cell are?
(A) $\frac{\pi}{6}r^3$ (B) $\frac{16}{3}\pi r^3$
(C) $\frac{16}{3}\pi^2 a^3$ (D) $\frac{\pi a^3}{3\sqrt{2}}$

Q5. A compound alloy of gold and copper crystallizes in a cubic lattice in which gold occupy the lattice point at corners of the cube and copper atom occupy the centres of each of the cube faces. What is the formula of this compound.
(A) $AuCu_6$ (B) AuCu
(C) $AuCu_3$ (D) $AuCu_2$

Q6. Diamond structure can be considered as ZnS (Zinc blend) structure in which each $Zn^{2+} = 4$ in alternate tetrahedral void and S^{2-} in cubic close pack arrangement is replaced by one carbon atom. If C–C covalent bond length in diamond is 1.5Å , what is the edge length of diamond unit cell (z = 8).
(A) 3.46 Å (B) 6.92 Å
(C) 1.73 Å (D) 3 Å

Q7. In a solid AB having NaCl type structure A atoms occupy the corner of the cubic unit cell and the face centered positions. If all the face-centered atom along one of the axes are removed, choose the correct options: (Atomic weight of A = 40), Atomic weight of B = 60, r_A = 1.4 Å, r_B = 2.8 Å
(A) The packing fraction is 0.579
(B) The density of the resultant solid is 1.008 g/cc
(C) The packing faction is 0.679
(D) The density of the resultant solid is 2.008 g/cc

Q8. Which of the following are correct with respect to body centred arrangement ?
(A) The spheres in the first layers of close packing are slightly opened up
(B) Each sphere in the scond layer is in contact with four spheres of the first layer and the third layer is similar to the first layer.
(C) The coordination number of the each sphere is 12.
(D) In bcc arrangement only 68% of the volume is occupied by the spheres.

Q9. KF crystallizes in the NaCl type structure. If the radius of K^+ ions 132 pm and that of F^- ion is 135 pm, what is the closet K–K distance ?
(A) Can not say (B) 534 pm
(C) 755 pm (D) 378 pm

Q10. Which one of the following schemes of ordering closed packed sheets of equal sized spheres generate close packed lattice.
(A) ABCABC (B) ABACABAC
(C) ABBAABBA (D) ABCBCABCBC

Q11. Select correct option/s
(A) ABABABAB.....in 3D is HCP
(B) ABCABCABC.....in 3D is CCP
(C) ABCABACABABC is close packing
(D) C.N. of both HCP and CCP atoms are 12

Q12. Select the correct statement about FCC (ABCAB...) structure.
(A) Distance between nearest octahedral void and tetrahedral void is $\frac{\sqrt{3}a}{4}$
(B) Distance between two nearest octahedral void is $\frac{a}{\sqrt{2}}$
(C) Distance between two nearest tetrahedral void is $\frac{\sqrt{3}a}{2}$
(D) Distance between layer A and B is $2r\sqrt{\frac{2}{3}}$

Q13. In the closest packing of atoms A (radius : r_a), the radius of atom B that can be fitted into tetrahedral voids is :-
(A) 0.155 r_a (B) 0.225 r_a
(C) 0.414 r_a (D) 0.732 r_a

Q14. Which of following is /are metallic solid.
(A) Fe (B) Mn (C) Ag (D) Zn

Q15. A cubic unit cell contains manganese ions at the corners and fluoride ions at the center of each edge. The radius of Mn ion is 0.65 Å and that of F^- ion is 1.36 Å.
(A) The empirical formula of compound is MnF_3
(B) The coordination number of the Mn ion is 6
(C) The edge length of the unit cell is 4.02 Å
(D) The density of the compound is 2.86 gm/cc

Q16. In A^+B^- ionic compound, radii of A^+ and B^- ions are 180 pm and 187 pm respectively. The crystal structure of this compound will be :-
(A) CsBr type (B) CsCl type
(C) ZnS type
(D) Similar to diamond

Q17. The melting point of RbBr is 682°C, while that of NaF is 988°C. The principal reason that melting point of NaF is much higher than that of RbBr is that –
(A) The two crystals are not isomorphous
(B) The molar mass of NaF is smaller than that of RbBr
(C) The internuclear distance ($r_c + r_a$) is greater for RbBr than that for NaF.
(D) NaF has greater lattice energy than that of RbBr

Q18. The mass of a unit cell of CsCl corresponds to :-
(A) 1 Cs^+ and 1 Cl^- (B) 1 Cs^+ and 6 Cl^-
(C) 4 Cs^+ and 4 Cl^- (D) 8 Cs^+ and 1 Cl^-

Q19. Choose the correct option/options -
(A) Two adjacent face centre atom doesn't touch each other in fcc unit cell because they are not nearest atom of face each other in fcc lattice.
(B) Number of nearest Na^+ ions of another Na^+ in Na_2O crystal will be 6.
(C) Minimum distance between two cubical voids in simple cube unit cell lattice will be a where a is length of edge of unit cell.
(D) By defects in solids density of solids either remains constant or decreases but it can never increase.

Q20. In which of following density of lattice increases?
(A) Schottky defect
(B) Frenkel defect
(C) Impurity defect
(D) Metal excess defect

Q21. Select the correct statement(s) -
(A) The ionic crystal of AgBr has Frenkel defect
(B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90°$, $\gamma = 120°$ is hexagonal
(C) In ionic compounds having Frenkel defect the ratio r^+/r^- is low
(D) The co-ordination number of Na^+ ion in NaCl is 6

Q22. Select the correct statement (s)

(A) In NaCl crystal, each ion has an octahedral coordination.

(B) The C.N. of an atom in hcp structure is 12.

(C) Diamagnetic materials are attracted by the magnetic fields.

(D) An octahedral hole is smaller than a tetrahedral hole.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 23 to 24

The solid are characterised by incompressibility, rigidity and mechanical strength. Solids have definite volume, shape, slow diffusion, law vapour pressure and possesses the unique property of being rigid. Solids can be classified as crystalline or amorphous on basis of the nature of order present in arrangement of their constituent particles.

Q23. Which of the following is not correct for amorphous solid :-

(A) Isotropic in nature

(B) Super cooled liquid

(C) Long range order

(D) Irregular Shape

Q24. Correct option for crystalline solid is :-

(A) Isotropic Nature

(B) Rough Surface on cutting

(C) Pseudo solids

(D) Sharp M.P.

Paragraph for Question No. 25 to 26

On the basis of classification of symmetry, the lattice have been devided into seven system. These can be grouped into 7 crystal systems. With 7 crystal system, there are total 14 bravais lattice possibles.

Q25. Which of following crystal system has/have all side length equal :-

(A) Cubic (B) Tetragonal

(C) Rhombohedral (D) Triclinic

Q26. Which of following crystal system has/have body centered bravais lattice

(A) Cubic (B) Tetragonal

(C) Monoclinic (D) Orthorhombic

Paragraph for Question No. 27 to 31

In haxagonal close packing second row of spheres are placed in depressions of first row whereas third row is vertically aligned with first, 2nd with fourth forming a pattern AB. AB.....The number of nearest atoms is called the coordination number. It has six fold axis of symmetry.

Q27. What is the coordination number of a central sphere in hcp.

(A) 8 (B) 12

(C) 6 (D) 4

Q28. If $\frac{r_+}{r_-}$ lies between 0.225 to 0.414, cation occupies

(A) tetrahedral void (B) octahedral void

(C) trigonal void (D) cubic void

Q29. The coordination number of octahedral void is -

(A) 8 (B) 4

(C) 6 (D) 3

Q30. Which of the following packing is most efficient.

(A) hcp (B) fcc

(C) A&Bboth (D) bcc

Q31. ABC.ABC ..... is called

(A) hcp (B) fccp

(C) bccp (D) simple cubic

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q32. Match list-I with list-II and select the correct answer by using the codes given below :

List-I (Shape)	**List-II (Radius ratio)**
(A) Planar triangle	(I) 0.732
(B) Square planar	(II) 0.225
(C) Body centered cubic	(III) 0.155
(D) Tetrahedral	(IV) 0.414

Codes :

	A	B	C	D
(A)	III	IV	I	II
(B)	III	II	I	IV
(C)	II	I	IV	III
(D)	I	III	IV	II

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q33. Match the column :

Column-I	**Column-II**
(A) Tetragonal and Hexagonal	(P) are two crystal systems
(B) Cubic and Rhombohedral	(Q) $a = b \neq c$
(C) Monoclinic and Triclinic	(R) $a \neq b \neq c$
(D) Cubic and Hexagonal	(S) $a = b = c$

Q34. Match the crystal system / unit cells mentioned in Column I with their characteristic features metioned in Column II.

Column-I	**Column-II**
(A) simple cubic and face-centred cubic	(P) have these cell parameters $a = b = c$ and $\alpha = \beta = \gamma$
(B) cubic and rhombohedral	(Q) are two crystal systems
(C) cubic and tetragonal	(R) have only two crystallographic angles of 90°
(D) hexagonal and monoclinic	(S) belong to same crystal system

Q35. Match the crystal system/unit cells mentioned in Column I with their characteristic features mentioned in Column II.

Column I	**Column II**
(A) Simple cubic and face-centered	(P) have these cellcubic parameters
(B) cubic and rhombohedral	(Q) are two crystal systems
(C) cubic and tetragonal	(R) have only two crystallo-graphic angles of 90°
(D) hexagonal and monoclinic	(S) belong to same crystal system

Q36. Match the column :

Column I	**Column II**
[Systems]	**[Packing fraction]**
(A) Simple cubic	(P) 0.74
(B) FCC	(Q) 0.68
(C) BCC	(R) 0.414
(D) hcp	(S) 0.52

Q37. Match the column :

Column I	**Column II**
(A) Rock salt structure	(P) Co-ordination number of cation is 4
(B) Zinc Blend structure	(Q) $\frac{\sqrt{3}a}{4} = r_+ + r_-$
(C) Flourite structure	(R) Co-ordination number of cation and anion are same
	(S) Distance between two nearest anion is $\frac{a}{\sqrt{2}}$

Q38. Match the column :

Column I	**Column II**
(A) If radius ratio $x\left(\frac{r_c}{r_a}\right) < 0.155$	(P) Coordination no. is 8
(B) If $0.225 \le x < 0.414$	(Q) Coordination no. is 4
(C) If $0.414 \le x < 0.732$	(R) Coordination no. is 6
(D) If $0.732 \le x < 1$	(S) Coordination no. is 2

Q39.

Column-I	**Column-II**
(A) H_2O(ice)	(P) Insulator
(B) Fe	(Q) Molecular Solid
(C) $C_{(graphite)}$	(R) Conductor
(D) ZnS (Solid)	(S) Metallic Solid
	(T) Covalent Solid

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q40. **Statement 1 :** Glass is isotropic
Statement 2 : Glass is supercooled liquid.

Q41. **Statement 1 :** Diamond is a covalent solid.
Statement 2 : The coordination number of each carbon atom in diamond is 4.

Q42. **Statement 1 :** Stoichiometric compounds obey the law of constant composition.
Statement 2 : Schottky and Frenkel defects are observed in stoichiometric compounds.

Q43. **Statement 1 :** An ionic structure is composed of oppositely charged ions.
Statement 2 : If the larger ions are close packed then the smaller ions may occupy either the octahedral holes or the tetrahedral holes depending on their size.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

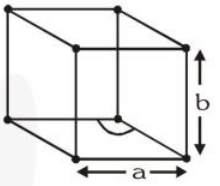
Q44. Number of solids, that are amorphous from given below polyurethane, Teflon, Potassium nitrate, Polyvinyle Chloride, Fibre glass, Copper.

Q45. Total number of bravais lattice of cubic crystal system is -

Q46. If C.N. in S.C. is x, C.N. is BCC is y and C.N. in FCC is Z then value of $\frac{x \times y}{z}$ is -

Q47. A solid crystallises in cubic crystal in which 'X' atom occupy all the corners and body centres and y atoms occupy all the face - centres and simplest formula of solid is X_aY_b then find value of (a + b).

Q48. Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined, the lattice constants were a = 4.53Å, and b = 7.60Å (see figure). How many molecules are contained in a given unit cell? [density (ice) = 0.92 gm/cm^3]



Q49. An element crystallizes in BCC structure. The edge of its unit-cell is 288 pm. If the density of the crystals is 7.2 g cm^{-3}, what is the atomic mass of the element (in gm/mole)?

Q50. The edge length of unit cell of a metal having molecular weight 75 g/mol is 5Å which crystallizes in cubic lattice. If the density is 2 gm/cm^3 then find the radius of metal atom. ($N_A = 6 \times 10^{23}$).
Give the answer in pm.

Q51. One unit cell of a mineral is composed of A^{+2} and C^{-2} and B^{+n}. A^{+2} ions at corners, C^{-2} ions at face centres and B^{+n} at the centre of the cube are present. What is the oxidation number (+n) of B^{+n} ion.

Q52. Calculate the number of atoms in a cubic based unit-cell having one atom on each corner and two atoms one each body diagonal.

Q53. Find the percentage void in HCP unit cell.

Q54. If the radius of Mg^{2+} ion, Cs^+ ion, O^{2-} ion, S^{2-} ion and Cl^- ion are 0.65 Å, 1.69 Å, 1.40 Å, 1.84 Å, and 1.81Å respectively. Calculate the sum of co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl.

Q55. Number of Na^+ present in per unit cell of Na_2O is?

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Amorphous substances show :
(1) short and long range order
(2) short range order
(3) long range order
(4) have no sharp m.p.
(A) 1 and 3 are correct
(B) 2 and 3 are correct
(C) 3 and 4 are correct
(D) 2 and 4 are correct

Q2. How many unit cells are present in 39 gm of potassium that crystallies in body centred cubic structure? [At.wt. of K = 39]

(A) 6.02×10^{23} (B) $\frac{6.02\times10^{23}}{4}$

(C) $\frac{6.02\times10^{23}}{2}$ (D) $\frac{3}{4}\times6.02\times10^{23}$

Q3. The crystal system of a compound with unit cell dimensions, a = 0.387, b = 0.387 and c = 0.504 nm, and $\alpha = \beta = 90°$ and $\gamma = 120°$ is :-
(A) Cubic
(B) Hexagonal
(C) Orthorhombic
(D) Rhombohedral

Q4. Which of the following are the correct axial distances and axial angles for rhombohedral system?
(A) $a = b = c$, $\alpha = \beta = \gamma \neq 90°$
(B) $a = b \neq c$, $\alpha = \beta = \gamma = 90°$
(C) $a \neq b \neq c$, $\alpha = \beta = \gamma = 90°$
(D) $a \neq b \neq c$, $\alpha \neq \beta \neq \gamma \neq 90°$

Q5. An oxide of 'A' crystallizes with eight atoms at the corners of the unit cell and 18 oxygen atoms on the edges & on the faces symmetrically. What is the formula of this oxide -
(A) AO_6 (B) A_2O_4
(C) A_3O_6 (D) A_3O_8

Q6. A solid compound contains X, Y and Z atoms in a cubic lattice with X atoms occupying the corners, Y atoms in the body centred positions and Z atoms at the centre of the faces of the unit cell. What is the empirical formula of the compound ?
(A) XY_2Z_3 (B) XYZ_3
(C) $X_2Y_2Z_3$ (D) X_8YZ_6

Q7. An alloy of metal 'A' 'B' and 'C' is found to have 'A' constituting ccp lattice. If 'B' atom occupy the edge centres and 'C' is present at the body centre, the formula of the alloy is :-
(A) A_4B_2C (B) A_4B_4C
(C) A_4B_3C (D) ABC

Q8. Packing fraction in 2-D hexagonal arrangement is -

(A) $\frac{\pi}{3\sqrt{2}}$ (B) $\frac{\pi}{3\sqrt{3}}$

(C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi}{6}$

Q9. For an ionic crystal of the general formula AX and coordination number 6, the values of radius ratio will be?
(A) > .73 (B) 0.41 < _ < 0.73
(C) 0.22 < _ < 0.41 (D) < 0.22

Q10. If the ratio of co-ordination number of A that of B is X : Y, then the ratio of number of atoms of A to that number of atoms of B in unit cell is -
(A) X : Y (B) Y : X
(C) X^2 : Y (D) Y : X^2

Q11. An element cystallizes in both fcc & bcc lattice. If the density of the element in the two forms is same, the ratio of unit cell edge length of fcc to that of bcc lattice is :-

(A) $(2)^{1/3}$ (B) $(1/2)^{1/3}$

(C) $(4)^{1/3}$ (D) $(1/4)^{1/3}$

Q12. When heated above 916°C, iron changes its bcc crystalline form to fcc without the change in the radius of atom. The ratio of density of the crystal before heating and after heating is :-

(A) 1.069 (B) 0.918

(C) 0.725 (D) 1.231

Q13. Sodium metal exists in bcc unit cell. The distance between nearest sodium atoms is 0.368 nm. The edge length of the unit cell is:

(A) 0.368 nm (B) 0.184 nm

(C) 0.575 nm (D) 0.425 nm

Q14. First three nearest neighbour distances for body centered cubic lattice are respectively (edge length of unit cell = a) :-

(A) $\sqrt{2}a, a\sqrt{3}a$ (B) $\frac{a}{\sqrt{2}}, a, \sqrt{3}a$

(C) $\frac{\sqrt{3}a}{2}, a, \sqrt{2}a$ (D) $\frac{\sqrt{3}a}{2}, a, \sqrt{3}a$

Q15. The number of nearest neighbours around each particle in a face-centred cubic lattice is :-

(A) 4 (B) 6

(C) 8 (D) 12

Q16. Platinum crystallizes in a face-centered cubic crystal with a unit cell edge length 'a'. The distance between nearest neighbour atom is :

(A) a (B) $a\frac{\sqrt{3}}{2}$

(C) $a\frac{\sqrt{2}}{2}$ (D) $a\frac{\sqrt{2}}{4}$

Q17. A crystal is made of particles X, Y and Z forms fcc packing. Y occupies all the octahedral voids of X and Z occupies all the tetrahedral voids of X. If all the partical along one body diagonal are removed then the formula of the crystal would be :-

(A) XYZ_2 (B) X_2YZ_2

(C) $X_8Y_4Z_5$ (D) $X_5Y_4Z_8$

Q18. A compound is formed by three element A, B and C. The element A form ccp, atoms of B occupy 2/3rd of tetrahedral void and atoms of c occupy 1/3rd of octrahedral voids then formula of the compound is :-

(A) A B C (B) $A_3B_4C_3$

(C) A_3B_4C (D) A_4B_3C

Q19. In a compound, atoms of element Y form ccp lattice and those of element X occupy 2/3rd of tetrahedral voids. The formula of the compound will be :-

(A) X_4Y_3 (B) X_2Y_3

(C) X_2Y (D) X_3Y_4

Q20. In a structure of mixed oxide, oxide ions are in CCP. One fifth of tetrahedral voids are occupied by divalent ion (A^{2+}) while half of the octahedral voids are occupied by trivalent ion (B^{3+}), the formula is :-

(A) $A_4B_5O_{10}$ (B) A_2BO_4

(C) AB_2O_4 (D) $A_5B_4O_{10}$

Q21. A certain oxide of metal M crystallises in such a way that O^{-2} ions occupy hcp arrangement following ABAB----------pattern. The metal ions however occupy 2/3rd of the octahedral voids then formula of the compound is –

(A) M_2O_3 (B) M_3O

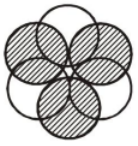
(C) MO_2 (D) M_3O_2

Q22. Octahedral void is surrounded by .....spheres :

(A) 6 (B) 4

(C) 8 (D) 12

Q23. The empty space between the shaded balls and hollow balls as shown in the diagram is called :-



(A) Hexagonal void
(B) octahedral void
(C) Tetrahedral void
(D) Double triangular void

Q24. A binary solid (A^+B^-) has a rock salt structure. If the edge length is 400 pm and radius of cation is 75 pm, the radius of anion is :-

(A) 100 pm (B) 125 pm
(C) 250 pm (D) 325 pm

Q25. KCl crystallizes in the same type of lattice as does NaCl. Given that $r_{Na^+} / r_{Cl^-} = 0.50$ and $r_{Na^+} / r_{K+} = 0.70$, calculate the ratio of the edge length of unit cell for KCl to that for NaCl :-

(A) 1.143 (B) 1.224
(C) 1.414 (D) None

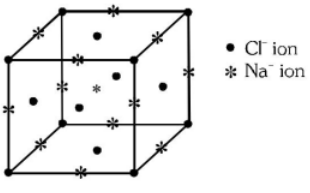
Q26. If the radius ratio is in the range of 0.414 - 0.732 then the co-ordination number will be :

(A) 2 (B) 4
(C) 6 (D) 8

Q27. The two ions A^+ and B^- have radii 88 and 200 pm respectively. In the closed packed crystal of compound AB, predict the co-ordination number of A^+.

(A) 6 (B) 8
(C) 4 (D) None

Q28.



In this NaCl type unit cell anion Cl^- adopts :-

(A) B.C.C.
(B) AB AB AB ..... type
(C) ABC ABC ABC .....type
(D) ABC CBA ABC CBA .....type

Q29. CaO and NaCl have the same crystal structure and nearly the same ionic radii. If x is the lattice energy of NaCl, the lattice energy of CaO is very nearly :-

(A) 2x (B) x
(C) 4x (D) $\frac{x}{2}$

Q30. In a compound XY the radius of cation X^+ and Y^- are 1.69Å and 1.81Å respectively. What will be the ratio of coordination numbers for X^+ and Y^-.

(A) 6 : 6 (B) 1 : 1
(C) 4 : 4 (D) 8 : 8

Q31. What type of crystal defect is indicated in the diagram below ?

Na^+ Cl^- Na^+ Cl^- Na^+ Cl^-
Cl^- □ Cl^- Na^+ □ Na^+
Na^+ Cl^- □ Cl^- Na^+ Cl^-
Cl^- Na^+ Cl^- Na^+ □ Na^+

(A) Frenkel defect
(B) Schottky defect
(C) Interstitial defect
(D) Frenkel and Schottky defects

Q32. NaCl is doped with 2×10^{-3} mol% $SrCl_2$. The concentration of cation vacancies is :

[Hint : Concentration of vacancies = $\frac{2\times10^{-3}}{100} N_A$]

(A) 3.01×10^{18} mol^{-1}
(B) 12.04×10^{18} mol^{-1}
(C) 6.02×10^{18} mol^{-1}
(D) 12.04×10^{20} mol^{-1}

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statement(s) is/are incorrect for NaCl lattice ?

(A) An Na^+ ion is placed at a distance (1/6)a directly above each Cl^- ion (where a is a cubic unit-cell edge length)

(B) An Na^+ ion is placed at a distance (1/4)a directly above each Cl^- ion

(C) An Na^+ ion is placed at a distance of $\frac{1}{2\sqrt{2}}$ a directly above each Cl^- ion

(D) An Na^+ ion is placed at a distance $\frac{1}{\sqrt{2}}$ a directly above each Cl^- ion

Q2. A cubic solid is made up of two elements P and Q. Atoms Q are present at the corners of the cube and atoms of P at the body centre. Which of the following is/are correct ?

(A) The formula of the compound is PQ

(B) The coordination number of P and Q are the same

(C) Coordination number of Q = 6

(D) Coordination number of P = 8

Q3. Which of the following statement(s) is/are correct –

(A) NaCl is a 'AB' crystal lattice that can be interpreted to be made up of two individual fcc unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.

(B) In a face centred cubic unit cell, the body centre is an octahedral void.

(C) In fcc unit cell, octahedral and tetrahedral voids are equal in number.

(D) Tetrahedral voids = 2 × octahedral voids, is valid for ccp and hcp.

Q4. Select the correct statement(s) with respect to Rock-Salt structure of ionic compounds :

(A) Co-ordination number of cation is four whereas that of anion is six.

(B) Co-ordination number of cation is six whereas that of anion is four.

(C) Co-ordination number of each cation & anion is four.

(D) Co-ordination number of each cation & anion is six.

Q5. In an AB unit crystal of NaCl types assuming Na^+ forming FCC

(A) The nearest neighbour of A^+ is $6B^-$ ion

(B) The nearest neighbour of B^- is 6 A^+ ion

(C) The second neighbour of A^+ is 12 A^+ ion

(D) The packing fraction of AB crystal is $\frac{\sqrt{3}\pi}{8}$

Q6. Which of the following statement (s) is/are correct?

(A) The coordination number of each type of ion in CsCl crystal is 8.

(B) A metal that crystallizes in bcc structure has a coordination no. of 12.

(C) A unit cell of an ionic crystal shares some of its ions with other unit cells.

(D) The length of the unit cell in NaCl is 552 pm (r_{Na^+} = 95pm, r_{Cl^-} = 181pm)

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 7 to 9

Density of a unit cell is represented as

$$\rho = \frac{\text{Effective no. of atoms (s)} \times \text{Mass of a unit cell}}{\text{Volume of a unit cell}}$$

$$= \frac{Z.M.}{N_A.a^3}$$

where, mass of unit cell = mass of effective no. of atoms (s) or ion (s).

M = At. wt./formula wt.

N_A = Avogadro's no. $\Rightarrow 6.023 \times 10^{23}$

a = edge length of unit cell

Q7. Silver crystallizes in a fcc lattice and has a density of 10.6 g/cm^3. What is the length of an edge of the unit cell ?

(A) 0.407 nm (B) 0.2035 nm

(C) 0.101 nm (D) 4.07 nm

Q8. An element crystallizes in a structure having fcc unit cell of an edge 200 pm. Calculate the density, if 100 g of this element contains 12×10^{23} atoms :

(A) 41.66 g/cm^3 (B) 4.166 g/cm^3

(C) 10.25 g/cm^3 (D) 1.025 g/cm^3

Q9. The density of KBr is 2.75 g/cm^{-3}. The length of the edge of the unit cell is 654 pm. To which type of cubic crystal, KBr belongs ?

(A) simple cubic (B) bcc

(C) fcc (D) none of these

Paragraph for Question No. 10 to 13

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (hcp), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assumed radius of every sphere to be 'r'.

Q10. The number of atoms in this hcp unit cell is -

(A) 4 (B) 6 (C) 12 (D) 17

Q11. The volume of this hcp unit cell is -

(A) $24\sqrt{2}r^3$ (B) $16\sqrt{2}r^3$

(C) $12\sqrt{2}r^3$ (D) $64r^3 / 3\sqrt{3}$

Q12. The empty space in this hcp unit cell is -

(A) 74% (B) 47.6%

(C) 32% (D) 26%

Q13. The arrangement in hcp unit cell is of which type?

(A) AAA.... (B) ABABAB....

(C) ABCABC.... (D) ABBAABBA...

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14. Match the column :

Column-I **[Bravais lattice(s)]**	**Column-II** **[Crystal system]**
(A) Primitive, face centered, body centered, end centered	(P) Cubic
(B) Primitive, face centered, body centered	(Q) Orthorhombic
(C) Primitive, body centered	(R) Hexagonal
(D) Primitive only	(S) Tetragonal

Q15.

Column-I	**Column-II**
(A) S.C	(P) % packing efficiency = 74%
(B) B.C.C	(Q) % packing efficiency = 52%
(C) F.C.C	(R) Coordination number = 8
(D) Monoclinic	(S) At least one angle ≠ 90°

Q16.

Column-I (Ionic Crystal)	**Column-II**
(A) CsCl	(P) $r + r^- = \frac{a\sqrt{3}}{4}$
(B) CaF_2	(Q) C.N (cation) = 4
(C) ZnS	(R) C.N (anion) = 8
(D) Na_2O	(S) C.N (anion) = 4

Q17. Match the column :

Column-I	**Column-II**
(A) Simple vacancy defect	(P) Shown by non-ionic solids and increases density of the solide
(B) Simple interstitial defect	(Q) Shown by ionic solids and decreases density of the solid
(C) Frenkel defect	(R) Shown by ionic solids and density of the solid decreses
(D) Schottky defect	(S) Shown by ionic solids and density of the solid remains the same

TRIPLE MATCH TYPE

*This section contains TWO tables (each having 3 columns and 4 rows). Based on each table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column - I	**Column - II**	**Column - III**
(P) Cubic	(I) a = b = c	(A) No. of Bravais lattice = 4
(Q) Hexagonal	(II) $\alpha = \beta = \gamma$	(B) No. of Bravais lattice = 1
(R) Rhombohedral	(III) $a \neq b \neq c$	(C) Body centered
(S) Orthorhombic	(IV) $\alpha \neq \beta \neq \gamma$	(D) Primitive

Q18. Incorrect combination is -

(A) P - I - D (B) P - II - C (C) R - II - B (D) S - II - B

Q19. Choose correct combination out of these :-

(A) P - I - A (B) R - I - B (C) S - III - B (D) Q - IV - B

Q20. In which of the following is correct for all 3 Column :-

(A) Q - II - B (B) Q - III - D (C) P - II - D (D) R - II - C

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q21. **Statement 1 :** ZnO becomes yellow when it is heated.

Statement 2 : NaCl becomes yellow when heated in the presence of Na vapours due to anion vacancy.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q22. Find the number of tetrahedral voids in hexagonal primitive unit cell.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	D	A	A	A	C	A	A	C	D
11	12	13	14	15	16	17	18	19	20
D	A	C	C	D	C	B	B	C	A
21	22	23	24	25	26	27	28	29	30
B	B	A	A	A	B	A	B	B	B
31	32	33	34	35	36	37	38	39	40
B	C	A	C	C	C	A	D	A	A
41	42	43	44	45	46	47	48	49	50
B	C	A	C	B	B	A	D	D	C
51	52	53	54	55	56	57	58	59	60
B	B	C	B	A	B	A	D	D	C
61	62	63	64						
D	B	D	B						

Ex-1A

1. C, D **2.** A, B, C **3.** A, C **4.** B, D **5.** C
6. A **7.** B, C **8.** A, B, D **9.** D **10.** A, B, D
11. A, B, C, D **12.** A, B, D **13.** B **14.** A, B, C, D **15.** A, B, C, D
16. A, B **17.** C, D **18.** A **19.** B, C **20.** C, D
21. B, D **22.** A, B **23.** C **24.** D **25.** A, C
26. A, B, D **27.** B **28.** A **29.** C **30.** C
31. B **32.** A
33. (A) → P, Q ; (B) → P,S ; (C) → P,R ; (D) → P
34. (A) P, S ; (B) -P,Q ; (C) - Q ; (D) - Q,R
35. (A) → P,S ; (B) → P,Q ; (C)→ Q ;(D)→ Q,R
36. (A) → S ; (B) → P ; (C)→ Q ;(D)→ P
37. (A) → R,S ; (B) → P,Q,R,S ; (C) → Q
38. (A) → S ; (B) → Q ; (C)→ R ; (D)→ P
39. (A) P, Q; (B) R, S; (C) R, T; (D) P
40. A **41.** B **42.** B **43.** B **44.** 4
45. 3 **46.** 4 **47.** 5 **48.** 4 **49.** 52
50. 2.16Å **51.** 4 **52.** 9 **53.** 26 **54.** 18
55. 8

Ex-2

1	2	3	4	5	6	7	8	9	10
D	C	B	A	A	B	C	C	B	B
11	12	13	14	15	16	17	18	19	20
A	B	D	C	D	C	D	C	A	A
21	22	23	24	25	26	27	28	29	30
A	A	B	B	A	C	A	C	C	D
31	32								
B	B								

Ex-2A

1. A, C, D **2.** A, B, D **3.** A, B, D **4.** D **5.** A, B, C
6. A, C, D **7.** A **8.** A **9.** C **10.** B
11. A **12.** D **13.** B
14. (A) → Q; (B) → P; (C) → S; (D) → R
15. (A) → Q; (B) → R; (C) → P; (D) → S
16. (A) - R; (B) - P,S; (C) - P,Q,S; (D) - P, Q, R
17. (A) → R ; (B) → P ; (C) → S ; (D) → Q
18. D **19.** B **20.** C **21.** B **22.** 12

-: Important Notes & Formulas :-